

Nirma University

Institute of Management

Trimester End Examination (IR/RPR), October-2022

Master of Business Administration, Trimester - IV (makeup)

MFT5SEEF16 Investment and Portfolio Management

c/10
M

Roll / Exam No.

Supervisor's initial
with date

Time: 3 Hours

Max. Marks: 100

Instructions:

1. Do NOT use pencils while answering.
2. All Calculations are part of your answer. Time Value tables should be given.
3. Write to the point.
4. Full marks will be awarded for neat and clean presentation.
5. Assumption/s if any, state clearly before answering the question.
6. Attempt all questions.
7. Figures to right in bracket indicate full marks.

Q-1. Mr. Nayan Sharma, a wealthy businessman, has approached you for professional advice on investment. He has a surplus of Rs. 30 lakhs which he wishes to invest in a stock market. Being risk averse by nature and a first timer to secondary market, he makes it very clear that the risk should be optimized. Having done some research in this field, you recommend to him a portfolio of two shares - stocks of an FMCG company AB Ltd. and an oil exploration company CD Ltd. You tell him that both are reputed companies. You have the following market data at your disposal. [30]

Period	Market return (%)	Returns (%)	
		AB Ltd.	CD Ltd.
1	22	27	12
2	24	24	18
3	18	18	21
4	-10	-18	20
5	-8	-11	-10
6	20	24	14

Mr. Sharma requests you to calculate the average returns, standard deviation, variance. He also asked you to calculate covariance, alpha and beta of both the company with respect to the market and estimate the CAPM of both the companies assuming the risk-free rate of 7%.

Q-2. A zero-coupon bond of Rs 100,000 has a term to maturity of six years and a market yield of 8 percent at the time of issue. [20]

- (A) What is the present value of bond?
- (B) What is the duration of the bond?
- (C) What is the modified duration of the bond?
- (D) What will be the percentage change in the price of the bond, if the yield declines by 0.3 percentage points (30 basis points?)

- Q-3. Consider the following information for three mutual funds, A, B, and C, and the market. [25]

Fund Schemes	Mean return (%)	Standard deviation (%)	Beta
A	16	21	1.7
B	13	12	0.9
C	19	16	1.2
Market index	14	13	1

The average risk-free rate was 7 percent. Evaluate performance of these funds and market using appropriate performance evaluation tools.

- Q-4. The current dividend on equity share of ABC Ltd. is Rs. 5. The company is expected to achieve 30% growth per year in next 5 years. Thereafter, the growth rate will fall and stabilise at 10%. Equity investors require 16% return. [25]

- What will be the intrinsic value of the stock?
 - What should be the intrinsic value of the stock if the 40% growth is expected decline to linearly to the constant growth of 10% over the period of 6 years?
 - What other aspects of company fundamentals that you will consider for better investment opportunities?
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